

Abelian Extensions Problem Set 1

Introductory

1. If L/K is an extension of fields, $\text{Aut}(L/K)$ is a group.
2. If $\alpha \in L$ is algebraic over K with minimal polynomial $m_{\alpha,K}(x)$, then $\sigma(\alpha)$ is a root of $m_{\alpha,K}$ for any $\sigma \in \text{Aut}(L/K)$.
3. If $L = K(\theta)$ with the minimal polynomial $m_{\theta,K}(x)$, then $|\text{Aut}(L/K)| \leq \deg(m_{\theta,K})$
4. If $d \in \mathbb{Z}$ is squarefree, then the function $\sigma : \mathbb{Q}(\sqrt{d}) \rightarrow \mathbb{Q}(\sqrt{d})$, $\sigma(a + b\sqrt{d}) = a - b\sqrt{d}$ is an automorphism, and $\sigma^2 = \text{id}$ is the identity automorphism.
5. $\text{Aut}(\mathbb{Q}(\sqrt[3]{5})/\mathbb{Q})$ is trivial
6. Let $n \in \mathbb{Z}^+$ and let $\zeta_n = e^{\frac{2\pi i}{n}} \in \mathbb{C}$. Then
 - ζ_n is a primitive n th root of unity.
 - The powers of ζ_n are precisely the n roots of $x^n - 1$.
7. $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \hookrightarrow (\mathbb{Z}/n\mathbb{Z})^\times$
8. Show: if L/K is abelian and F/L is abelian, then F/K need not be abelian.

Advanced

1. If L_1/K and L_2/K are finite Galois extensions, then the composite field L_1L_2/K is Galois over K .
2. Under the same hypotheses, $\text{Gal}(L_1L_2/K) \hookrightarrow \text{Gal}(L_1/K) \times \text{Gal}(L_2/K)$. Conclude that $[L_1L_2 : K] \leq [L_1 : K][L_2 : K]$ and that if $\text{Gal}(L_1/K)$ and $\text{Gal}(L_2/K)$ are both abelian, then $\text{Gal}(L_1L_2/K)$ is abelian as well.